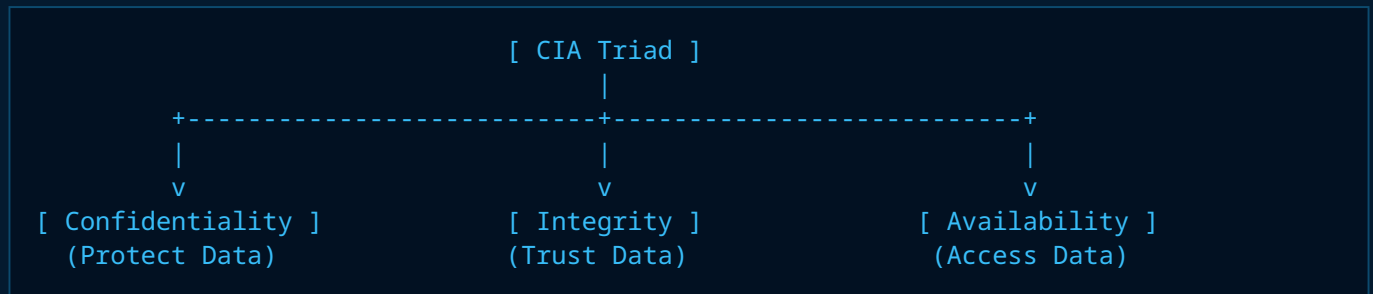


[CIA Triad (Cybersecurity Fundamental)]

The CIA Triad is one of the most important concepts in cybersecurity. It represents the 3 main goals of security:

1. Confidentiality
2. Integrity
3. Availability

It helps organizations protect data, systems, and networks.



[1. Confidentiality]

[Meaning]

Only authorized people should access the data. Unauthorized users should NOT see sensitive information.

[Real-Life Example]

- Your bank password should only be known by you.
- Employee salary data should only be accessible by HR.
- Medical records should only be viewed by doctors and patients.

[Threats to Confidentiality]

- Phishing attacks
- Password theft
- Insider threats
- Data leaks
- Malware
- Packet sniffing

[Methods to Protect Confidentiality]

- Strong passwords
- Multi-Factor Authentication (MFA)
- Encryption

- Access control
- VPN
- Data classification

[SOC Analyst Perspective]

A SOC Analyst monitors:

- Unauthorized login attempts
- Suspicious file access
- Privilege escalation
- Data exfiltration alerts
- Failed login spikes

[2. Integrity]

[Meaning]

Data should remain accurate and unchanged. No unauthorized modification should happen.

[Real-Life Example]

- Bank transaction amounts should not change.
- Log files should not be tampered with.
- Exam results should remain accurate.

[Threats to Integrity]

- Malware
- SQL Injection
- Insider modification
- Log tampering
- Man-in-the-middle attacks

[Methods to Protect Integrity]

- Hashing
- Digital signatures
- File integrity monitoring
- Access permissions
- Backups
- Version control

[SOC Analyst Perspective]

A SOC Analyst checks:

- Unexpected file changes
- Log tampering
- Suspicious database activity

- Unauthorized configuration changes
 - Integrity alerts from SIEM/EDR
-

[3. Availability]

[Meaning]

Systems and data should be accessible when needed. Authorized users should get services without interruption.

[Real-Life Example]

- A company website should stay online.
- Employees should access email anytime.
- Banking apps should work properly.

[Threats to Availability]

- DDoS attacks
- Ransomware
- Hardware failure
- Server crash
- Power outage

[Methods to Protect Availability]

- Backups
- Disaster recovery plan
- Redundancy
- Load balancing
- UPS power systems
- Monitoring systems

[SOC Analyst Perspective]

A SOC Analyst monitors:

- Server downtime alerts
 - DDoS traffic spikes
 - Ransomware activity
 - Service failures
 - CPU/memory spikes
-

[Easy Table for Revision]

CIA Part	Meaning	Main Goal	Example
Confidentiality	Prevent unauthorized access	Privacy	Password protection

CIA Part	Meaning	Main Goal	Example
Integrity	Prevent unauthorized changes	Accuracy	Hash verification
Availability	Ensure systems stay accessible	Uptime	Backup servers

[Easy Daily-Life Analogy]

Imagine a locker:

- **Confidentiality** → Only you have the key
- **Integrity** → Nobody changes items inside
- **Availability** → You can open it anytime

[Important Interview Point]

Many interviewers ask:

“Explain CIA Triad with example.”

Simple answer:

CIA Triad stands for Confidentiality, Integrity, and Availability. Confidentiality protects data from unauthorized access, Integrity ensures data accuracy, and Availability ensures systems are accessible when needed.

[Short SOC L1 Summary]

- **Confidentiality:** Protect data from unauthorized access.
- **Integrity:** Protect data from unauthorized modification.
- **Availability:** Ensure systems and services remain accessible.

One-Line Memory Trick:

CIA = Protect Data, Trust Data, Access Data